

CMPT 225-D100 Quiz 2, Summer 2026

Duration	30 minutes for the entire exam.
Coding Questions	For coding questions, use C++-like pseudocode. This means that small coding errors will usually be overlooked. However, it must be clear that the code could easily be converted to real C++.
Aids allowed	Pencil or pen, and eraser. No notes, no papers, no books, no computers, no calculators, no cheat sheets, etc.
During the exam	Raise your hand if you would like to speak with a proctor. To be fair to all students, we don't answer questions about the exam. Choose the answer based on your best understanding of the

question. Re-reading the question may help.

Quiz 2

Question 1

Suppose you are given a stack *S* containing 0 or more ints, and can only run the essential stack operations on it. *S.size()* is **not** considered an essential operation for this question. Write **C++-like pseudocode** that calculates and then prints the sum of the elements in *S*. After the sum is printed, *S* should have the same values, in the same order, as before the code was called. You can use 0 or more extra stacks to help.

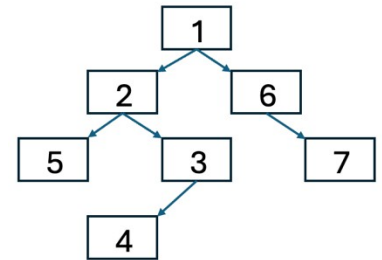
Important: only uses stacks in your answer! No arrays, lists, vectors, queues, etc.

```
sum = 0 // init sum
temp = make_empty_stack() // make empty temp stack
while (not S.empty()) {
    val = S.pop() // remove the top element of S
    sum += val // add it to the sum
    temp.push(val) // push it onto the other stack
}
print(sum) // print sum
while (not temp.empty()) { // put elements of temp back onto S
    S.push(temp.pop())
}
```

Question 2

a) Write the numbers of the binary tree on the right in **preorder** order.

1, 2, 5, 3, 4, 6, 7



b. Write the numbers of the binary tree on the right in **inorder** order.

5, 2, 4, 3, 1, 6, 7

c. Write the numbers of the binary tree on the right in **postorder** order.

5, 4, 3, 2, 7, 6, 1

d. Write a **C++ struct** (use exact C++!) that defines a node of a binary tree.

```
struct Node {  
    int data;  
    Node* left;  
    Node* right;  
};
```

e. Write **C++-like pseudocode** that, given a pointer to the root of any binary tree of ints (with nodes as defined in the previous question), returns the sum of the ints in the tree (an empty tree has sum 0). For example, for the tree above has sum 28.

```
int treeSum(p) {  
    if (p == null) return 0;  
    return node->value + treeSum(p->left) + treeSum(p->right)  
}
```

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Solutions